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Key Points:

- A 3-D P-wave velocity model of Sulawesi obtained from body wave tomography reveals an inter-linked network of slabs and partial melt
- Potential magma reservoirs are detected beneath volcanoes at the eastern end of the North Arm of Sulawesi above the Sangihe and Halmahera slabs
- A diffuse low velocity anomaly beneath the solitary Colo volcano may be a consequence of decompression melting arising from extension

Supporting Information:

Supporting Information may be found in the online version of this article.

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Evidence of Magma Reservoirs Beneath Volcanoes in Northern Sulawesi and the Molucca Sea From Regional Earthquake Tomography

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Abstract We present a new 3-D P-wave velocity model of the crust and upper mantle beneath Sulawesi from regional earthquake body wave tomography. The arrival time data were sourced from the Agency for Meteorology, Climatology, and Geophysics (BMKG) Indonesian earthquake catalog for the period 2018 through to 2023, which is based on records from 126 seismic stations in Sulawesi and its neighborhood. Our results reveal clear evidence of high velocity subducting slabs beneath the Molucca Sea Collision Zone and northern Sulawesi, along with low velocity zones above the Sangihe and Halmahera slabs, that likely represent magma reservoirs. Furthermore, we found a diffuse low-velocity anomaly in the mantle beneath Colo volcano, a solitary island volcano located in the distant back-arc region of the North Sulawesi subduction zone, which is consistent with decompression melting induced by extension.

Plain Language Summary This study presents a new and detailed 3-D model of seismic wavespeed variations in the crust and upper mantle beneath the island of Sulawesi in Indonesia. Data from over 6,000 earthquakes that occurred between 2018 and 2023 and were recorded by 126 seismic stations distributed throughout Sulawesi were exploited to construct the model using a method called seismic tomography. The results reveal the presence of subduction zones in northern Sulawesi and the Molucca Sea, in which one tectonic plate descends into the mantle beneath another as they converge. Low velocity anomalies in the uppermost mantle beneath northern Sulawesi correspond with the location of active volcanoes at the surface, and therefore may represent accumulations of partial melt. One significant finding is a diffuse low-velocity zone beneath Colo volcano, the only active volcano south of the North Sulawesi subduction zone, which may represent decompression melting caused by extension.

1. Introduction

The Island of Sulawesi lies to the east of Borneo, and is surrounded by the Celebes sea to the north, the Makassar Strait to the west, the Banda Sea to the south and the Molucca Sea to the east. It essentially consists of a patchwork of island arcs, ophiolites and continental fragments from both Eurasia and Gondwana that were assembled primarily as a consequence of the rapid northward motion of the Australian continent over the last ~45 Myr (Hall et al., 2011). Its unmistakable “K” shape (see Figure 1) comprises four arms that have distinct geological histories.

In the north, the current tectonic configuration is dominated by the North Sulawesi subduction zone that converges at a rate of 42–50 mm/year (Socquet et al., 2006), with southerly subduction of the Celebes Sea slab beginning around 8 Ma (Hall et al., 2011). The Slab 2 model of Hayes et al. (2018) indicates that a seismically deforming slab extends to a depth of ~200 km. Recent adjoint waveform tomography of eastern Indonesia (Wehner, Blom, et al., 2022; Wehner, Rawlinson, et al., 2022) suggests that the slab may penetrate down to the top of the transition zone with a near vertical dip, after bending in the top ~100–150 km, which may explain the concentration of shallower seismicity. One of the more unusual aspects of the North Sulawesi subduction zone is the complete absence of Neogene or active volcanism, despite the presence of a south-dipping Celebes slab and the 100 km Benioff zone contour approximately aligning with the southern coast of the north arm (Hall, 2018). Volcanic activity is observed only at the easternmost end of the north arm, and is associated with the west-dipping

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Sangihe slab and east dipping Halmahera slab (Hanyu et al., 2012). The sole volcano, Colo, is located on the island Una-Una in Gorontalo Bay, some 100 km offshore of the north arm, and arguably to the south of the deepest penetration of the Celebes Sea slab (Cottam et al., 2011; Hall, 2018). Furthermore, the cluster of seismicity in the neighborhood of Una-Una is relatively shallow, and characterized by normal faulting, which is consistent with a crustal thinning/extensional origin to the volcanism (Serhalawan & Chen, 2024). Indeed, Pholbud et al. (2012) show that rapid subsidence in Gorontalo Bay (where Colo volcano sits) without significant faulting of the basin, is likely due to low-angle normal faulting driven by subduction roll-back at the North Sulawesi trench, which is consistent with the presence of young core complexes and detachments on land (Spencer, 2011).

In the center of Sulawesi, active deformation is mainly controlled by the left-lateral strike-slip Palu-Koro Fault, which has a slip-rate of 42 mm/year (Socquet et al., 2006) and was the source of the 2018 (Mw 7.5) destructive earthquake and tsunami (Supendi, Nugraha, Widiyantoro, Pesicek, et al., 2020). Recent geodetic modeling results (Nijholt et al., 2024) confirm that interplay exists between the shallow subduction interface of the Celebes Sea slab and the Palu-Koro Fault, with observed surface motions best explained by slow slip on the subduction interface in combination with deep afterslip on the Palu-Koro Fault. This subduction-transform edge propagator setting likely influences the seismogenic behavior of both the strike-slip and subduction segments.

Recent global body wave tomography by Hall and Spakman (2015) reveals that subducted slabs beneath the northern and eastern parts of Sulawesi and the Molucca Sea Collision Zone (MSCZ) have complex geometries. The MSCZ is situated in the middle of the Molucca Sea, between the eastern Halmahera microplate and the west Sangihe microplate. As a result of their collision, a high ridge was formed in the middle of the two microplates, which is known as the Talaud-Mayu Ridge (Hamilton, 1979). The Molucca Sea Plate (MSP) lies between the southern Philippines and northeastern Indonesia, and subducts to the east beneath the Halmahera Arc and to the west beneath the Sangihe Arc (Fang et al., 2022). Seismic anisotropy in the northern Sulawesi region was analysed by Cao et al. (2024), revealing mantle wedge corner flow in the Sangihe subduction zone. In the Gulf of Tomini, they found that trench-oblique fast-axis orientations indicate interactions between two subducting slabs, with the southernmost region showing E–W fast directions in the mantle wedge, possibly related to Sula slab subduction.

Body wave tomography conducted by previous studies in Sulawesi and its neighborhood (Hall & Spakman, 2015; Widiyantoro & van der Hilst, 2007; Zenonos et al., 2019, 2020) were restricted by sparse station coverage and few local events, which meant that resolution of the uppermost mantle and crust was severely limited. Since the Mw7.5 Palu Earthquake of 2018, Sulawesi has been much more densely covered by permanent stations of the Indonesian Seismic Network, thus allowing many more local earthquakes to be detected and providing more crossing paths for improved tomographic resolution (see Figure S1 in Supporting Information S1). This has greatly increased the potential of body wave tomography to recover structural detail in key regions, including subduction zones and beneath active volcanoes. The goal of this study is to better understand the configuration of multiple overlapping subducting plates in the Sulawesi region, and how this might influence magmatic plumbing systems beneath active volcanoes.

2. Data and Methods

In this study, we use data from the BMKG earthquake catalog between 1 January 2018 and 30 July 2023, which is based on recordings from 126 BMKG permanent broadband stations (see the blue inverted triangles in Figure 1a). A total of 19,031 events of magnitude 2.0–7.5 were detected during this period with 418,358 identified P- and 79,685 S-wave arrival times. All arrival time picks were made by professional seismic analysts from BMKG, who preferentially target P-wave arrivals for event location purposes. Consequently, the associated S-wave data set is sparse (<20% of the total number of P-picks), which limits its usefulness in the following analysis.

We use an iterative non-linear tomographic inversion for variations in V_p , V_s , and hypocenter locations by applying the software package FMTOMO (Pilia et al., 2013; Rawlinson & Sambridge, 2004, 2005). FMTOMO uses the Fast Marching Method (De Kool et al., 2006; Sethian & Popovici, 1999) for the forward step of travel-time prediction in which the eikonal equation is solved on a grid of points. The main advantages of this method are robustness in the presence of extreme heterogeneity and computational efficiency. FMTOMO uses a regularized gradient based subspace inversion scheme to solve the linearized inversion step to constrain velocity parameters, and a fully non-linear grid search for earthquake location, which achieves computational efficiency by taking

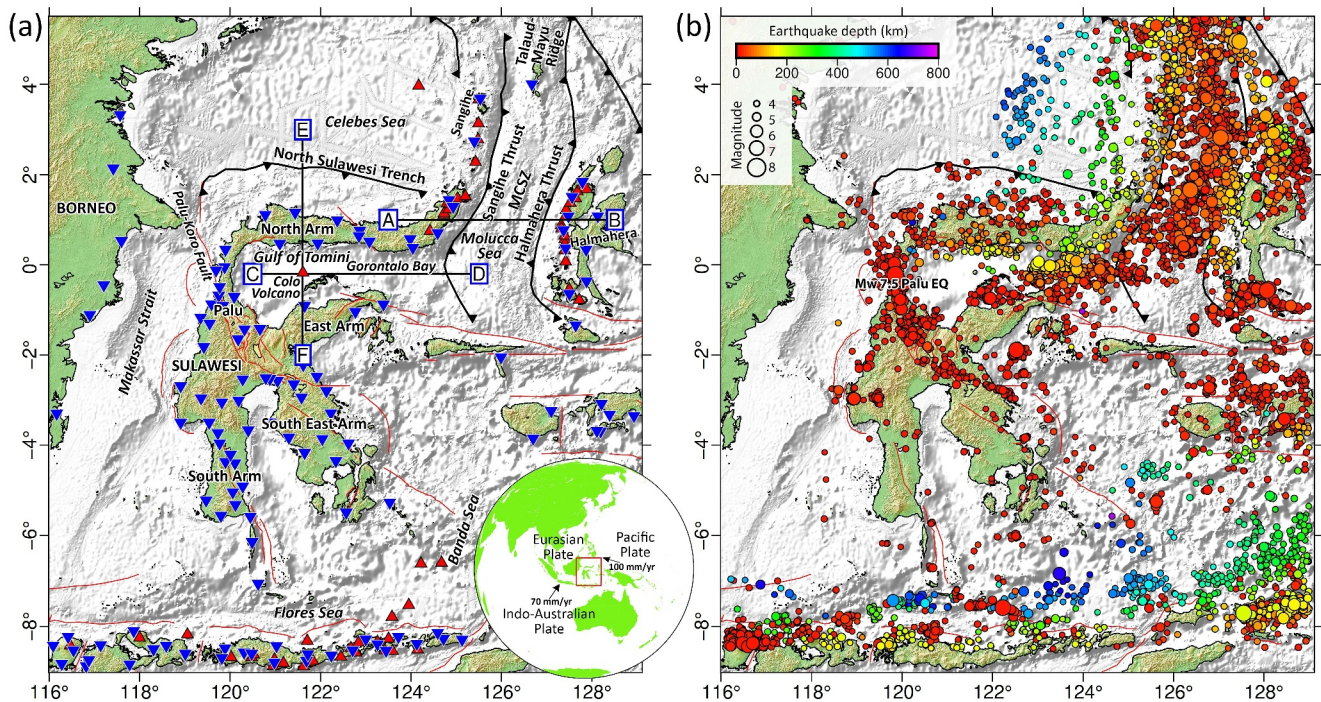


Figure 1. (a) Map of the study area, with the inset map showing its location in the Indonesian region. Blue inverted triangles show the BMKG seismic stations. Red triangles denote active Quaternary volcanoes. Lines A-B, C-D, and E-F depict the locations of the vertical sections shown in Figure 3. (b) Epicenters of earthquakes between January 2018 and July 2023 as a function of focal depth (color) and magnitude (size), as used in this study. The fault and trench data (red and black lines) were taken from Irsyam et al. (2020). Note that the dips of the Sangihe and Halmahera thrusts beneath the Molucca Sea have opposite polarity to their associated subducting slabs.

advantage of the grid-based nature of the eikonal solver (see Pilia et al., 2013 for more details). The sequential application of the forward and inversion steps iteratively solves the non-linear problem, since ray path geometries (derived from the calculated traveltimes) and source-receiver traveltimes are updated after each application of the subspace inversion scheme and grid search. The 3-D model is parameterized using a grid of control points, with cubic B-spline functions applied to generate a smooth velocity continuum.

The first step in our workflow is data preparation. We converted the BMKG earthquake catalog picks to the input format required by FMTOMO. The initial number of earthquakes available in the catalog is 19,031 with 418,358 associated P-picks and 79,685 S-picks. However, prior to inversion, we implement a selection criteria to eliminate poorly constrained events. The selection criteria are: (a) each event must be recorded by at least 10 seismic stations, (b) the earthquake magnitude must be ≥ 4.0 , (c) the azimuthal gap must be less than 210° for stations in the study area, and (d) travel-time residuals (differences between observations and starting model predictions) must be less than 2 s (the effect of this choice is discussed in Section 3). A total of 6,207 events (Figure 1b) with 106,722 P-picks and 16,916 S-picks passed the selection process and were subsequently used in the inversion (see Figures S1 and S2 in Supporting Information S1 for path coverage and traveltimes curves). The geographic and depth range of the model is 6°N – 9°S ; 115.7°E – 129.3°E and 5 km above sea level to 850 km below sea level, respectively (see Figure 1a). We use a grid spacing of 22 km for latitude, 19 km for longitude, and 19 km for depth.

Due to the underdetermined nature of the inverse problem, regularization is necessary in order to produce a stable and robust solution. The damping and smoothing regularization approach we use favors parsimonious models that are close to the starting model. In this study, the damping factor was empirically determined by using the trade-off between data variance and model variance, whereas the smoothing factor was empirically determined by using the trade-off between data variance and model roughness. The appropriate damping and smoothing values were chosen such that data variance is as small as possible while also minimizing model variance and roughness (i.e., in the neighborhood of the point of maximum curvature on the trade-off curves). Based on the application of this method and several inversion tests, the appropriate damping and smoothing parameters are 50 and 125

respectively for V_p , and 50 and 80 respectively for V_s (see Figures S3 and S4 in Supporting Information S1). V_p and V_s anomalies are computed with reference to the 1-D velocity model AK135 (Kennett et al., 1995), which provides a good initial fit to the data (see Figures S5 and S6 in Supporting Information S1). In this study, we do not attempt to draw inferences from V_p/V_s due to the relative sparsity of S-wave data, and our primary interpretation is largely based on the V_p results.

3. Results and Discussion

3.1. Resolution Test

To investigate the robustness of the recovered anomalies, we carried out two sets of checkerboard recovery tests with different grid sizes. The anomaly size in the smaller and larger checkerboard is twice and four times the size of the velocity grid spacing used in the real data inversion respectively. Maximum positive and negative perturbations of ± 0.15 km/s relative to the 1-D reference velocity model were used as inputs (with the same grid spacing as was used in the inversion of arrival time observations). A spike test focusing on the two slow-velocity anomalies (0.08 km/s reduction) is also carried out. The associated synthetic traveltimes data set in each case was generated by the forward solver in FMTOMO.

If the inversion results from the synthetic data—which use the same regularization as the model produced from the observations—are able to recover the corresponding input anomaly, even with a reduction in amplitude due to the implementation of regularization, we consider that anomaly to be resolved by the data. This provides a means of deciding which regions of the model can be interpreted with confidence, while keeping in mind the limitations of checkerboard tests (Rawlinson & Spakman, 2016). We conducted several synthetic tests to analyse how the individual addition of noise and relocation affects the checkerboard recovery. The results of the checkerboard tests are summarized in Figures S7–S12 in Supporting Information S1. It is clear from these tests that structural recovery beneath much of North Sulawesi and the MSCZ is sufficient to permit interpretation in the upper 200 km of the model, while limited recovery and some smearing can be observed below 200 km depth, although the larger anomaly sizes exhibit improved recovery at greater depths, which is to be expected. Furthermore, the spike test results (Figure S13 in Supporting Information S1) indicate that the two anomalies are able to be recovered using the data set and method in this study in the case of 0.2 s standard deviation of added noise. The effects of relocation and noise on the recovery of velocity structure are minimal, as indicated in Figures S7–S10 in Supporting Information S1.

3.2. V_p Model and Constraints on Anomaly Amplitude

After three iterations, the data misfit has converged (Figure S14 in Supporting Information S1) and the hypocenters locations stabilize (Figures S15 and S16 in Supporting Information S1). The final tomography model is illustrated as a series of key depth sections in Figure 2 and a set of vertical cross-sections in Figure 3. Given the geographic location of the study region, interpretation will focus on high velocity subducting slabs and low velocity zones beneath volcanic regions, which may be a signature of partial melt and/or elevated temperatures. In this context, it is important to explore the extent to which we can constrain anomaly amplitude. Figures 2 and 3 suggest that anomalies tend not to exceed 1.0%–1.5%, which in isolation, would make an interpretation of partial melt harder to justify, given that melt sources in the upper mantle beneath volcanic regions tend to have negative V_p anomalies as large as 5% or more (e.g., Supendi, Nugraha, Widiyantoro, Abdullah, et al., 2020). However, our data is sourced from the BMKG earthquake archive, in which picks are not extensively checked, resulting in errors that can be significant. While there is some data redundancy, path coverage can also be quite variable. In order to mitigate against these effects, any initial traveltimes residuals larger than 2°s were removed, and damping and smoothing regularization were applied to suppress model features not required by the data. Since these choices are ad hoc, we have investigated their effects on our results, with a particular focus on the amplitude of anomalies. For instance, if we reduce damping and smoothing to very low levels (0.1 in both cases), then the shallow low velocity zones are still present, but have amplitudes of up to 5% (see Figure S17 in Supporting Information S1). While this test indicates that much larger amplitudes are consistent with the data, we choose to present the more regularized model in the manuscript, since it was favored in the trade-off curve test, and contains fewer apparent artifacts, particularly at depth. However, it needs to be understood that the pattern of anomalies in the regularized model shown in Figures 2 and 3 is more robust than their amplitudes.

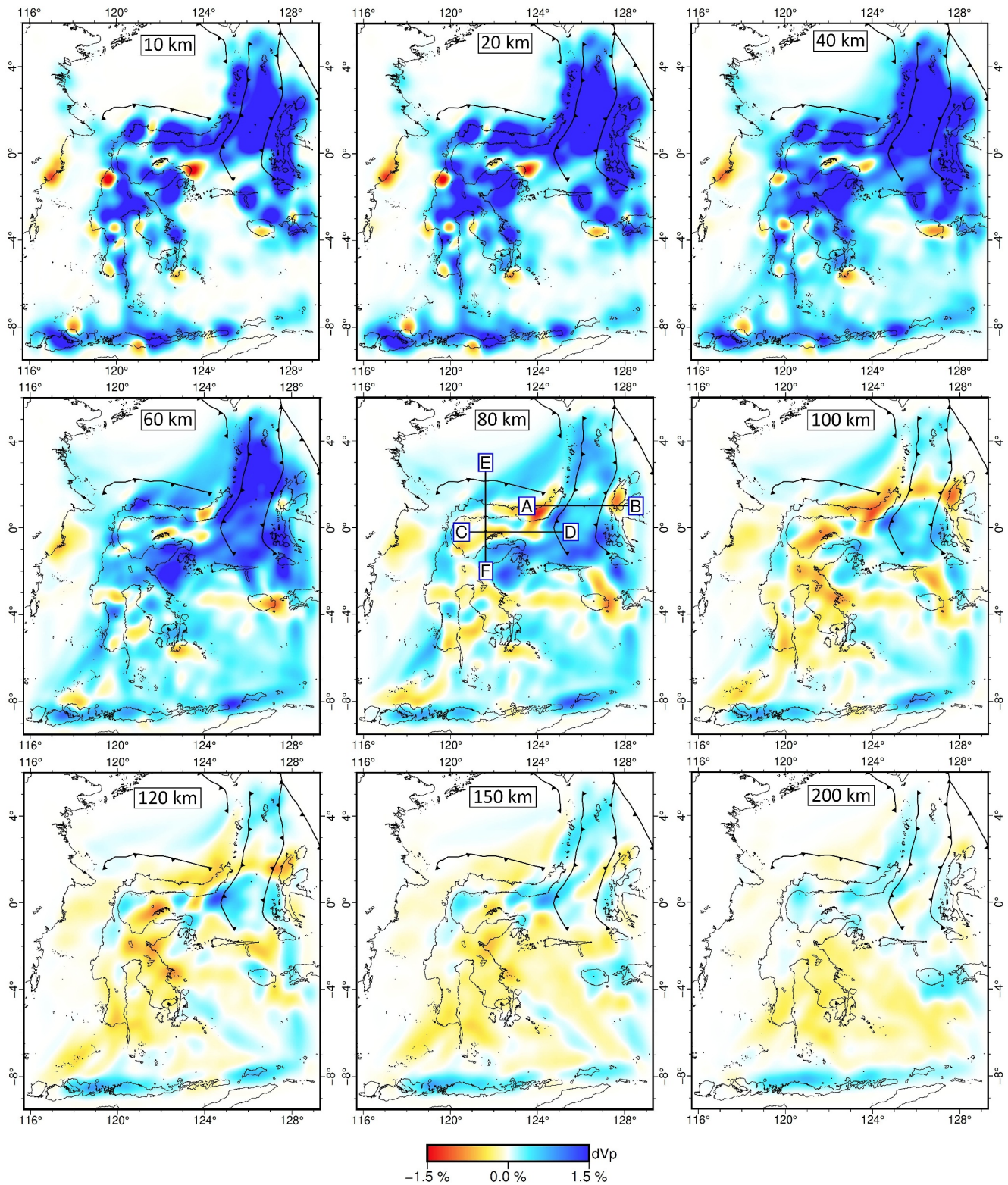


Figure 2. V_p models at 10, 20, 40, 60, 80, 100, 120, 150, and 200 km depth. Maps are plotted with a perturbation scale of $\pm 1.5\%$. Positive and negative anomalies are indicated by blue and red colors, respectively. Cross-sections through the velocity model in Figure 3 are located according to the black lines superimposed on the 80 km depth section.

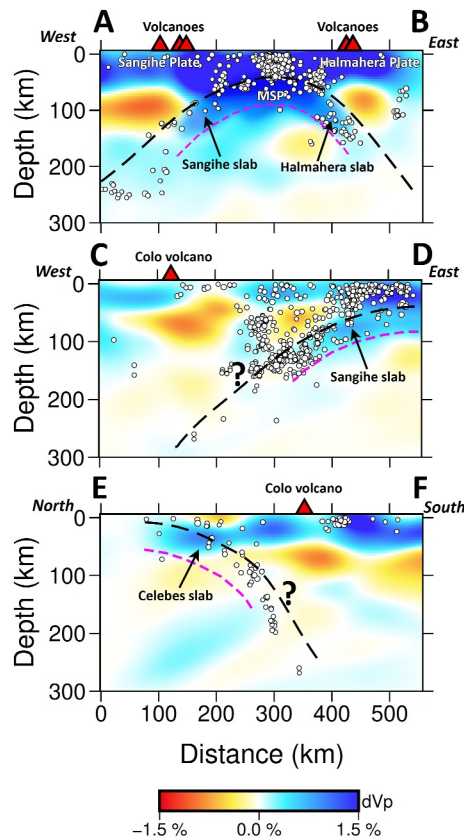


Figure 3. Vertical sections through the V_p model along A-B, C-D, and E-F (see Figure 1 or two for slice locations) with interpretations superimposed. Models are plotted with a perturbation scale of $\pm 1.5\%$ relative to $ak135$. The dashed black line defines the top of the slabs taken from the slab 2.0 model of Hayes et al. (2018), with question marks added beyond about 100 km in C-D and E-F to reflect uncertainty in geometry and location. The dashed magenta line is the inferred location (based on seismicity and velocity variations) of the bottom of the slab.

velocity anomaly beneath several volcanoes on islands in North Sulawesi and Molucca at a depth of ~ 80 – 130 km, which are likely to be zones of partial melt that serve as the magma source for crustal magma reservoirs (Kim et al., 2005).

As in section A-B (Figure 3), the high-velocity anomaly beneath the Molucca Sea, which is continuous to 200 km depth, is inferred to be the subducted slab of the MSP. In section C-D of Figure 3, the continuity of the MSP is less clear below 150 km depth, but this is likely a due to resolution dropping off with depth (see Figure S10 in Supporting Information S1). Section E-F of Figure 3 reveals a high-velocity anomaly down to 60 km depth and is likely related to the southward-subducting Celebes Sea slab. However, resolution is largely lost below this depth due to an absence of deep earthquakes further north, which explains why the slab in the uppermost mantle is not imaged. The Slab 2 model of Hayes and our new results are largely consistent in the upper 100–150 km of the mantle, and the location of shallow low velocity anomalies, which are interpreted as melt, are also consistent with the depth range (80–120 km) of the underlying slabs in terms of fluid release and melting (e.g., Syracuse et al., 2010). However, we do acknowledge that melt is not the only explanation, and that temperature and subsolidus effects (Takei, 2017) may also contribute to the amplitude and footprint of low velocity anomalies. At greater depths, the precise location and geometry of the slabs is uncertain. As a consequence, we have added question marks to the dashed lines marking out slab geometry (from the Slab 2 model) in the deeper regions of the model (Figure 3).

We detect a low velocity anomaly beneath Colo volcano in the Gulf of Tomini on both sections C-D and E-F in the depth range 60–110 km, which may be related to an underlying melt source. Colo volcano is Quaternary in age

We also performed a spike test with -5% input anomalies at the locations of two key low velocity anomalies with very low damping and smoothing (0.1 in each case) and 0.2 s standard deviation of added noise added to the synthetic data set (see Figure S18 in Supporting Information S1). The results show that our two input anomalies can be recovered, although inspection of the recovered spike anomalies suggest that their amplitude is approximately 2% or 3%, which is half the magnitude of the input anomalies. Finally, we also performed several tests in which the 2 s traveltimes residual filter is relaxed. Figure S19 in Supporting Information S1 shows the results when a 5 s traveltimes residual filter is used (i.e., any initial traveltimes residual larger than 5 s is removed from the inversion), and Figure S20 in Supporting Information S1 shows the result when a 1.5 s post-inversion filter is used (i.e., all arrival times are used as input, but after the inversion, all residuals between the observations and final model predictions greater than 1.5 s are removed, and the inversion is run again). The results are similar to those shown in Figures 2 and 3, although the anomaly amplitudes are larger, and the subducting slabs are less visible. The upshot of all these tests is that, while we favor interpretation of the pattern of anomalies shown in Figures 2 and 3, it is likely that their amplitudes are suppressed, which is a typical outcome of regularized inversion schemes (e.g., Rawlinson & Spakman, 2016).

3.3. Interpretation

Section A-B of Figure 3 depicts a continuous high-velocity anomaly that dips east and west down to 200 km depth (and possibly beyond, though we have limited resolution below 200 km depth), and is likely a signature of divergent double subduction (Zhang et al., 2017) of the MSP, which subducts westward beneath Sulawesi's north arm and the Sangihe Islands, and eastward beneath the Molucca Islands with a dip of $\sim 45^\circ$. This high-velocity zone is consistent with the distribution of the earthquake hypocenters, which confirms the presence of MSP subduction. Subducting slabs are generally characterized by high-velocity anomalies, which are caused by colder material descending into a hotter mantle (Zhao, 2015). Their geometry is typically delineated by earthquakes that rupture the descending crust, which is more rigid than the surrounding mantle (Hasegawa et al., 1978). Furthermore, we detected a low-

and has been the subject of much debate, with several different tectonic models postulated to explain its formation for example, the volcano is related to the southeastward dipping dormant subduction zone in the Celebes Sea (Katili, 1975); it is part of an extinct volcano located in Togean islands, which has been shifted to the north (Katili & Sudradjat, 1984); it is due to East Sangihe Subduction zone, which produced a Quaternary volcanic belt in the eastern part of the north arm that has extended to the southwest (Effendi & Bawono, 1997); it is caused by subduction of the Celebes Sea plate beneath the North Arm of Sulawesi at a shallow angle (Sendjaja, 2013); or it is a product of localized melting caused by extension, crustal thinning and delamination (Hall & Spakman, 2015). Fang et al. (2023) suggested that magma migrated to Colo volcano due to compression at either end of the North Sulawesi Trench, which moved southward as the Banggai–Sula macroblock pushed northward. Colo volcano is separate from Indonesia's volcanic belt that stretches from the north end of Sumatera-Java-Nusa Tenggara-Molucca to North Sulawesi; this area is complex and involved three plates (Celebes sea plate, MSP, and the micro continent Banggai Sula) that subducted and obducted between the Miocene and Quaternary period (Satyana et al., 2011).

A comprehensive study of the volcanic products from Colo has not yet been conducted. Vroon and van Bergen (2000) reported the presence of adakites from Colo and noted Sr and Nd isotopic ratios that differ from those of the Sangihe and Halmahera arc volcanics. They argued that these isotopic differences support the conclusion that the Colo volcanics are associated with the southward-subducting Celebes slab. If this hypothesis were to be true, it raises the question of why Colo volcano is located so far from the volcanic arc, and is either not above a subducting slab, or above one that is too deep to solicit melting in the overlying mantle. The unusual position of this single volcano therefore requires further explanation. Based on our tomography result (see section C-D and E-F in Figure 3), Colo volcano sits above a relatively diffuse low velocity zone that is different in character to the more pronounced low velocity zones immediately above the Sangihe slab. If decompression melting caused by extension were to be invoked to explain the existence of Colo volcano, then this observation would appear to be consistent with such a conclusion, given that such a mechanism tends to produce a relatively diffuse low velocity anomaly (e.g., Forsyth et al., 1998).

The corresponding S-wave tomography results, including checkerboard tests, are shown in Figures S21–S24 in Supporting Information S1. While it is clear from the checkerboard tests that the reliability of the model is inferior to that of the P-wave model, resolution in the upper 100 km is tolerable to the extent that a first-order comparison with the P-wave model is possible. It is therefore reassuring to see that the low velocity anomalies beneath the volcanoes, including Colo, are also present in S-wave model. However, there is little sign of velocity anomalies associated with the subducting slabs.

4. Conclusions

Local-regional earthquake data recorded by the BMKG network have permitted us to image several important structural features beneath Sulawesi, especially above 200 km depth, using body wave arrival time tomography. We have detected the presence of a high-velocity anomaly beneath the Molucca Sea that is consistent with divergent double subduction of the MSP. Furthermore, we suggest that low-velocity anomalies in the uppermost mantle directly above the west and east dipping slabs could indicate the presence of partial melt and/or fluids that drive active arc volcanism. Additional insights into partial melt distribution above the subducting slabs could be obtained through the study of shear wave velocity and attenuation. Our results also shed new light on the origins of the only volcano (Colo) in the back-arc region of the North Sulawesi subduction zone. Previous studies provide strong evidence of significant subsidence in the center of Gorontalo Bay, which is associated with the formation of a core complex on land, driven by extensional tectonics. This suggests that extension-driven melting may play a crucial role in the region, and the diffuse low velocity anomaly we image is arguably more consistent with such a source. However, further investigation is needed to properly understand the origin and magmatic plumbing system of Colo Volcano.

Data Availability Statement

The BMKG earthquake catalog picks are available at <https://repogempa.bmkg.go.id>. The FMTOMO package is freely available to download from Rawlinson (2006) and <https://github.com/nrawlinson/FMTOMO>. Data visualization was performed using Generic Mapping Tools version 6 (Wessel et al., 2019).

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References

- Cao, L., Lü, C., He, X., Rawlinson, N., Hao, T., Widiyantoro, S., et al. (2024). Mantle flow induced by the interplay of downgoing slabs revealed by seismic anisotropy beneath the Sula Block in eastern Indonesia. *Journal of Geophysical Research: Solid Earth*, 129(5), e2023JB028110. <https://doi.org/10.1029/2023JB028110>
- Cottam, M. A., Hall, R., Forster, M. A., & Boudagher-Fadel, M. K. (2011). Basement character and basin formation in Gorontalo Bay, Sulawesi, Indonesia: New observations from the Togian Islands. *Geological Society, London, Special Publications*, 355(1), 177–202. <https://doi.org/10.1144/SP355.9>
- De Kool, M., Rawlinson, N., & Sambridge, M. (2006). A practical grid-based method for tracking multiple refraction and reflection phases in three-dimensional heterogeneous media. *Geophysical Journal International*, 167(1), 253–270. <https://doi.org/10.1111/j.1365-246X.2006.03078.x>
- Effendi, A. C., & Bawono, S. S. (1997). *Geological map of the Menado sheet North Sulawesi scale 1:250,000*. Geological Research and Development Centre.
- Fang, G., Zhang, J., Hao, T., Dong, M., Jiang, C., & He, Y. (2022). The causal mechanism of the Sangihe Forearc Thrust, Molucca Sea, northeast Indonesia, from numerical simulation. *Journal of Asian Earth Sciences*, 237, 105350. <https://doi.org/10.1016/j.jseas.2022.105350>
- Fang, G., Zhang, J., Jiang, C., Dong, M., Hao, T., Li, Z., et al. (2023). Dynamic analysis of geophysical characteristics of Una-Una Volcanic Island, Sulawesi, Indonesia. *Journal of Asian Earth Sciences*, 255, 105767. <https://doi.org/10.1016/j.jseas.2023.105767>
- Forsyth, D. W., Scheirer, D. S., Webb, S. C., Dorman, L. M., Orcutt, J. A., Harding, A. J., et al. (1998). Imaging the deep seismic structure beneath a mid-ocean ridge: The MELT experiment. *Science*, 280(5367), 1215–1218. <https://doi.org/10.1126/science.280.5367.1215>
- Hall, R. (2018). The subduction initiation stage of the Wilson cycle. *Geological Society, London, Special Publications*, 470(1), 415–437. <https://doi.org/10.1144/SP470.3>
- Hall, R., Cottam, M. A., & Wilson, M. E. J. (2011). The SE Asian gateway: History and tectonics of Australia-Asia collision. *Geological Society, London, Special Publications*, 355, 1–6. <https://doi.org/10.1144/SP355.1>
- Hall, R., & Spakman, W. (2015). Mantle structure and tectonic history of SE Asia. *Tectonophysics*, 658, 14–45. <https://doi.org/10.1016/j.tecto.2015.07.003>
- Hamilton, W. B. (1979). Tectonics of the Indonesian region (Report No. 1078). <https://doi.org/10.3133/pp1078>
- Hanyu, T., Gill, J., Tatsumi, Y., Kimura, J., Sato, K., Chang, Q., et al. (2012). Across- and along-arc geochemical variations of lava chemistry in the Sangihe arc: Various fluid and melt slab fluxes in response to slab temperature. *Geochemistry, Geophysics, Geosystems*, 13(10), 2012GC004346. <https://doi.org/10.1029/2012GC004346>
- Hasegawa, A., Umino, N., & Takagi, A. (1978). Double-planed deep seismic zone and upper-mantle structure in the Northeastern Japan Arc. *Geophysical Journal International*, 54(2), 281–296. <https://doi.org/10.1111/j.1365-246X.1978.tb04260.x>
- Hayes, G., Moore, G. L., Portner, D. E., Hearne, M., Flamme, H., Furtney, M., & Smoczyk, G. M. (2018). Slab2—A comprehensive subduction zone geometry model. *U.S. Geological Survey*, 362(6410), 58–61. <https://doi.org/10.5066/F7PV6JNV>
- Irsyam, M., Cummins, P. R., Asrurifak, M., Faizal, L., Natawidjaja, D. H., Widiyantoro, S., et al. (2020). Development of the 2017 national seismic hazard maps of Indonesia. *Earthquake Spectra*, 36(1_suppl), 112–136. <https://doi.org/10.1177/8755293020951206>
- Katili, J. A. (1975). Volcanism and plate tectonics in the Indonesian island arcs. *Tectonophysics*, 26(3–4), 165–188. [https://doi.org/10.1016/0040-1951\(75\)90088-8](https://doi.org/10.1016/0040-1951(75)90088-8)
- Katili, J. A., & Sudradjat, A. (1984). The devastating 1983 eruption of Colo volcano, Una Una Island, central Sulawesi, Indonesia. *Geologisches Jahrbuch*, A75, 27–47.
- Kennett, B. L. N., Engdahl, E. R., & Buland, R. (1995). Constraints on seismic velocities in the Earth from traveltimes. *Geophysical Journal International*, 122(1), 108–124. <https://doi.org/10.1111/j.1365-246X.1995.tb03540.x>
- Kim, K.-H., Chiu, J.-M., Pujol, J., Chen, K.-C., Huang, B.-S., Yeh, Y.-H., & Shen, P. (2005). Three-dimensional V_p and V_s structural models associated with the active subduction and collision tectonics in the Taiwan region. *Geophysical Journal International*, 162(1), 204–220. <https://doi.org/10.1111/j.1365-246X.2005.02657.x>
- Nijholt, N., Simons, W., Riva, R., Efendi, J., Sarsito, D., & Broerse, T. (2024). Triggered and recurrent slow slip in North Sulawesi, Indonesia. *Tectonophysics*, 885, 230416. <https://doi.org/10.1016/j.tecto.2024.230416>
- Pholbud, P., Hall, R., Advokaat, E., Burgess, P., & Rudyawan, A. (2012). A new interpretation of Gorontalo Bay, Sulawesi. In *Proceedings Indonesian Petroleum Association, 36th Annual Convention, IPA12-G-039 1-23*.
- Pilia, S., Rawlinson, N., Direen, N. G., Cummins, P. R., & Balfour, N. (2013). Structural controls on localized intraplate deformation and seismicity in Southern Australia: Insights from local earthquake tomography of the Flinders Ranges. *Journal of Geophysical Research: Solid Earth*, 118(5), 2176–2190. <https://doi.org/10.1002/jgrb.50168>
- Rawlinson, N. (2006). FMTOMO: July 25, 2007 Release (version 1.0) [Software]. <https://nickrawlinson.com/fmtomo>
- Rawlinson, N., & Sambridge, M. (2004). Multiple reflection and transmission phases in complex layered media using a multistage fast marching method. *Geophysics*, 69(5), 1338–1350. <https://doi.org/10.1190/1.1801950>
- Rawlinson, N., & Sambridge, M. (2005). The fast marching method: An effective tool for tomographic imaging and tracking multiple phases in complex layered media. *Exploration Geophysics*, 36(4), 341–350. <https://doi.org/10.1071/EG05341>
- Rawlinson, N., & Spakman, W. (2016). On the use of sensitivity tests in seismic tomography. *Geophysical Journal International*, 205(2), 1221–1243. <https://doi.org/10.1093/gji/ggw084>
- Satyana, A. H., Faulin, T., & Mulyati, S. N. (2011). Tectonic evolution of Sulawesi area: Implications for proven and prospective petroleum plays.
- Sendjaja, P. (2013). *Petrology and geochemistry of volcanic rocks from the Togeian Islands Gulf of Tomini Central Sulawesi Province: Implication for the tectonic setting of Sulawesi Island* (Ph.D. Thesis). Institut Teknologi Bandung.
- Serhalawan, Y., & Chen, P.-F. (2024). Seismotectonics of Sulawesi, Indonesia. *Tectonophysics*, 883, 230366. <https://doi.org/10.1016/j.tecto.2024.230366>
- Sethian, J. A., & Popovici, A. M. (1999). 3-D traveltimes computation using the fast marching method. *Geophysics*, 64(2), 516–523. <https://doi.org/10.1190/1.1444558>
- Socquet, A., Simons, W., Vigny, C., McCaffrey, R., Subarya, C., Sarsito, D., et al. (2006). Microblock rotations and fault coupling in SE Asia triple junction (Sulawesi, Indonesia) from GPS and earthquake slip vector data. *Journal of Geophysical Research*, 111(B8). <https://doi.org/10.1029/2005JB003963>
- Spencer, J. E. (2011). Gently dipping normal faults identified with Space Shuttle radar topography data in central Sulawesi, Indonesia, and some implications for fault mechanics. *Earth and Planetary Science Letters*, 308(3–4), 267–276. <https://doi.org/10.1016/j.epsl.2011.06.028>

- Supendi, P., Nugraha, A. D., Widiyantoro, S., Abdullah, C. I., Rawlinson, N., Cummins, P. R., et al. (2020). Fate of forearc lithosphere at arc-continent collision zones: Evidence from local earthquake tomography of the Sunda-Banda arc transition, Indonesia. *Geophysical Research Letters*, 47(6), e2019GL086472. <https://doi.org/10.1029/2019GL086472>
- Supendi, P., Nugraha, A. D., Widiyantoro, S., Pesicek, J. D., Thurber, C. H., Abdullah, C. I., et al. (2020). Relocated aftershocks and background seismicity in eastern Indonesia shed light on the 2018 Lombok and Palu earthquake sequences. *Geophysical Journal International*, 221(3), 1845–1855. <https://doi.org/10.1093/gji/ggaa118>
- Syracuse, E. M., Van Keken, P. E., & Abers, G. A. (2010). The global range of subduction zone thermal models. *Physics of the Earth and Planetary Interiors*, 183(1–2), 73–90. <https://doi.org/10.1016/j.pepi.2010.02.004>
- Takei, Y. (2017). Effects of partial melting on seismic velocity and attenuation: A new insight from experiments. *Annual Review of Earth and Planetary Sciences*, 45(1), 447–470. <https://doi.org/10.1146/annurev-earth-063016-015820>
- Vroon, P. Z., & van Bergen, M. J. (2000). Geochemical evidence for two subducting plates beneath North Sulawesi (Indonesia). In *Abstract: Goldschmidt 2000, Oxford, UK* (p. 1060).
- Wehner, D., Blom, N., Rawlinson, N., Daryono, Böhm, C., Miller, M. S., et al. (2022). SASSY21: A 3-D seismic structural model of the lithosphere and underlying mantle beneath Southeast Asia from multi-scale adjoint waveform tomography. *Journal of Geophysical Research: Solid Earth*, 127(3), e2021JB022930. <https://doi.org/10.1029/2021JB022930>
- Wehner, D., Rawlinson, N., Greenfield, T., Daryono, Miller, M. S., Supendi, P., et al. (2022). SASSIER22: Full-waveform tomography of the eastern Indonesian region that includes topography, bathymetry, and the fluid ocean. *Geochemistry, Geophysics, Geosystems*, 23(11), e2022GC010563. <https://doi.org/10.1029/2022GC010563>
- Wessel, P., Luis, J. F., Uieda, L., Scharroo, R., Wobbe, F., Smith, W. H. F., & Tain, D. (2019). The generic mapping tools version 6. *Geochemistry, Geophysics, Geosystems*, 20(11), 5556–5564. <https://doi.org/10.1029/2019GC008515>
- Widiyantoro, S., & van der Hilst, R. (2007). Mantle structure beneath Indonesia inferred from high-resolution tomographic imaging. *Geophysical Journal International*, 130(1), 167–182. <https://doi.org/10.1111/j.1365-246X.1997.tb00996.x>
- Zenonos, A., De Siena, L., Widiyantoro, S., & Rawlinson, N. (2019). P and S wave travel time tomography of the SE Asia-Australia collision zone. *Physics of the Earth and Planetary Interiors*, 293, 106267. <https://doi.org/10.1016/j.pepi.2019.05.010>
- Zenonos, A., De Siena, L., Widiyantoro, S., & Rawlinson, N. (2020). Direct inversion of *S - P* differential arrival times for ratio in SE Asia. *Journal of Geophysical Research: Solid Earth*, 125(5), e2019JB019152. <https://doi.org/10.1029/2019JB019152>
- Zhang, Q., Guo, F., Zhao, L., & Wu, Y. (2017). Geodynamics of divergent double subduction: 3-D numerical modeling of a Cenozoic example in the Molucca Sea region, Indonesia. *Journal of Geophysical Research: Solid Earth*, 122(5), 3977–3998. <https://doi.org/10.1002/2017JB013991>
- Zhao, D. (2015). *Multiscale seismic tomography*. Springer.